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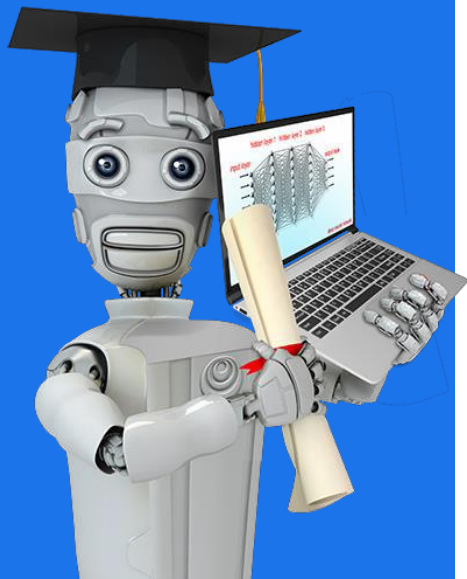
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Classification

Motivations

Classification

Question

Answer " y "

Is this email spam?

no yes

Is the transaction fraudulent?

no yes

Is the tumor malignant?

no yes

y can only be one of **two** values

"**binary** classification"

$class = category$

false true

0

1

*useful for
classification*

"negative class"

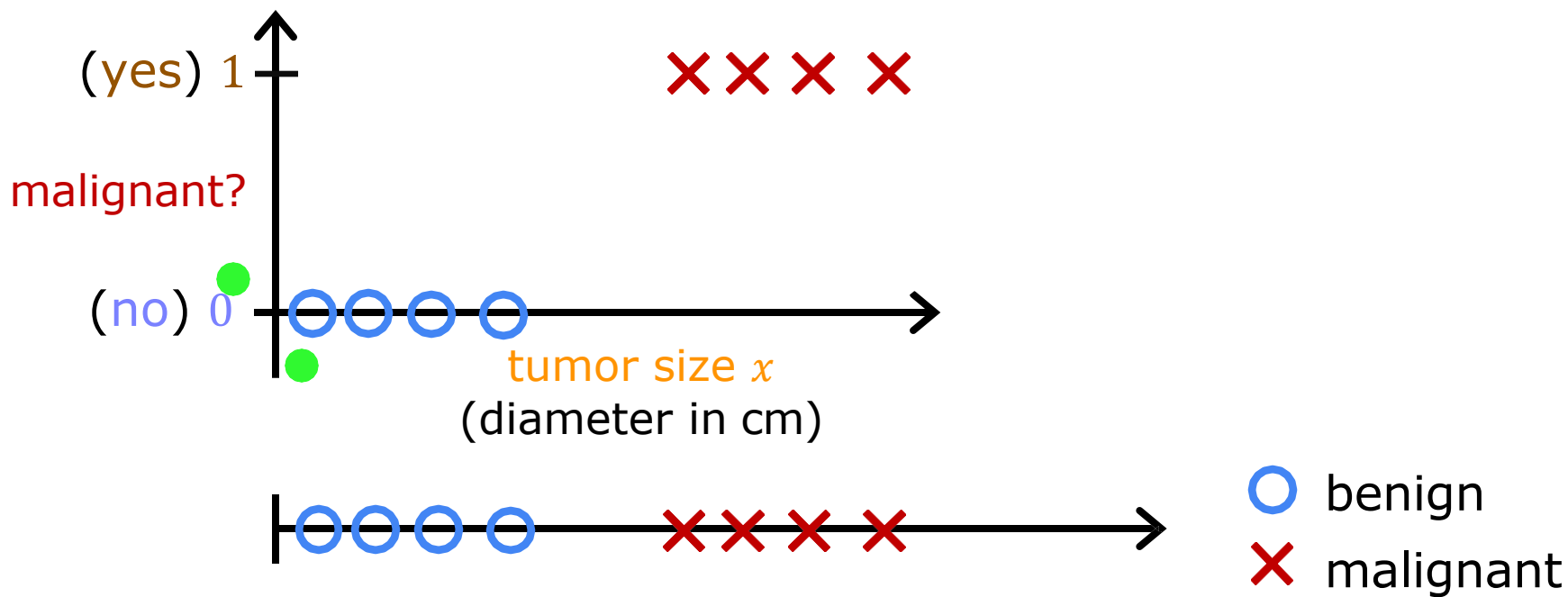
$\neq$ "bad"

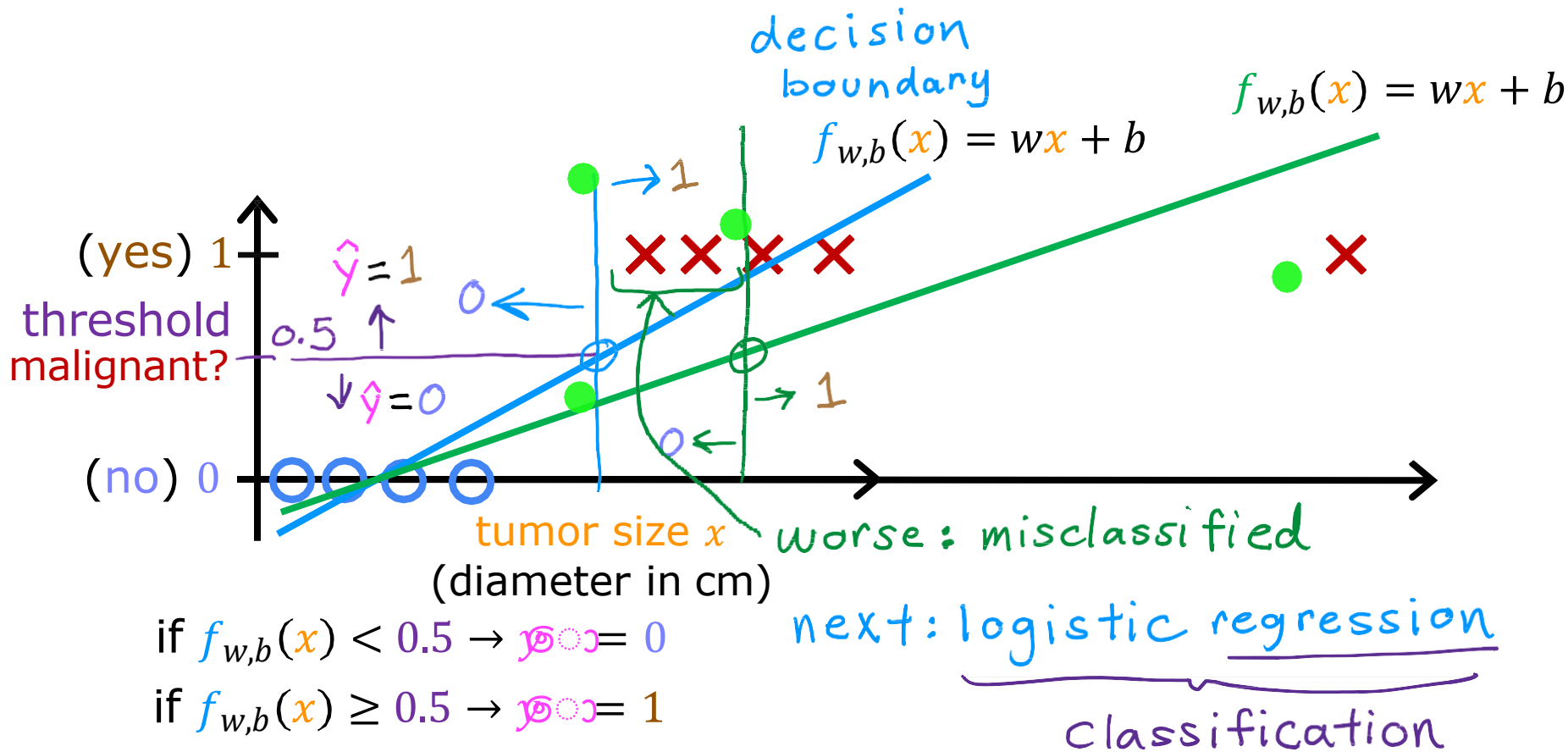
absence

"positive class"

$\neq$ "good"

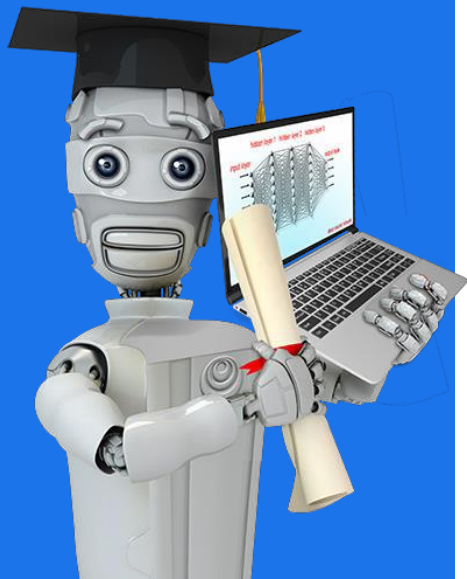
presence





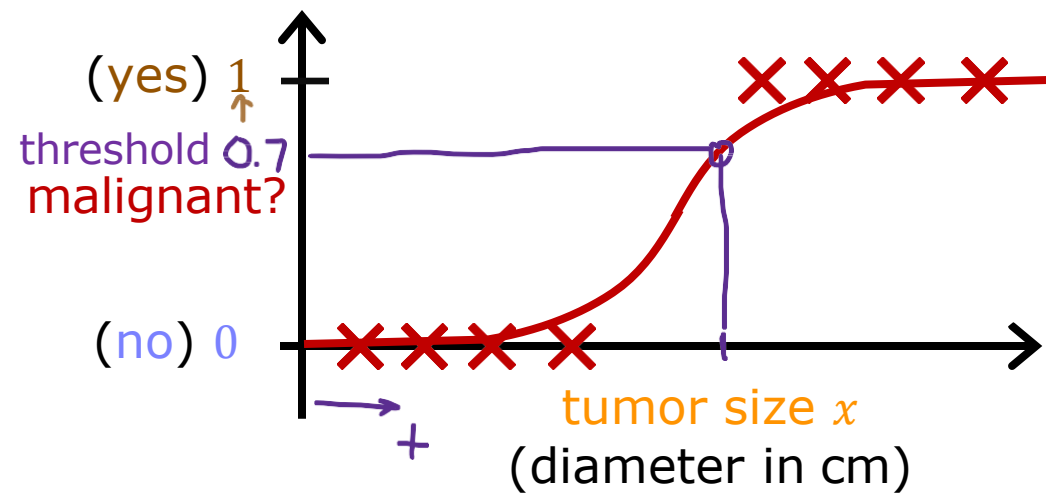
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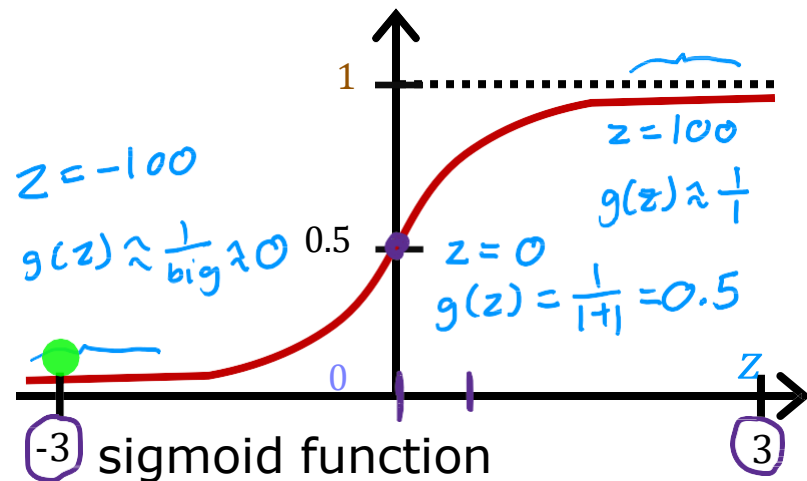


Classification

Logistic Regression



Want outputs between 0 and 1

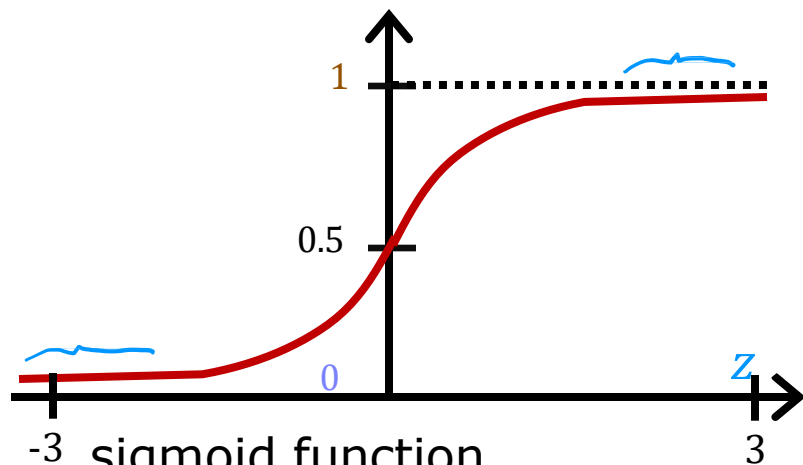


logistic function

outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \quad 0 < g(z) < 1$$

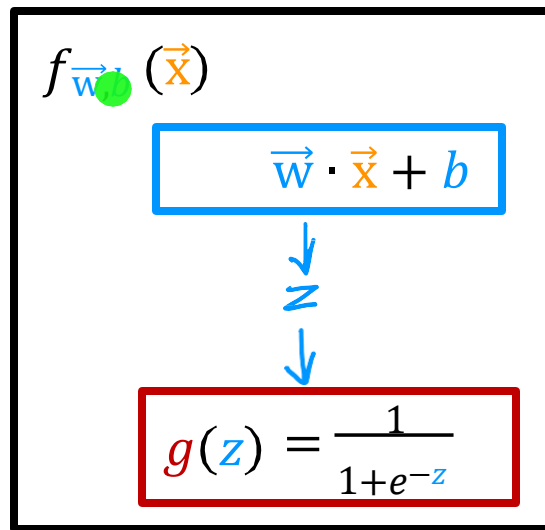
Want outputs between 0 and 1



logistic function

outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \quad 0 < g(z) < 1$$



$$f_{\vec{w}, b}(\vec{x}) = g(\underbrace{\vec{w} \cdot \vec{x} + b}_z) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

"logistic regression"

$e \approx 2.7$

Interpretation of logistic regression output

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

“probability” that class is 1

Example:

x is “tumor size”

y is 0 (not malignant)

or 1 (malignant)

$$f_{\vec{w},b}(\vec{x}) = 0.7$$

70% chance that y is 1

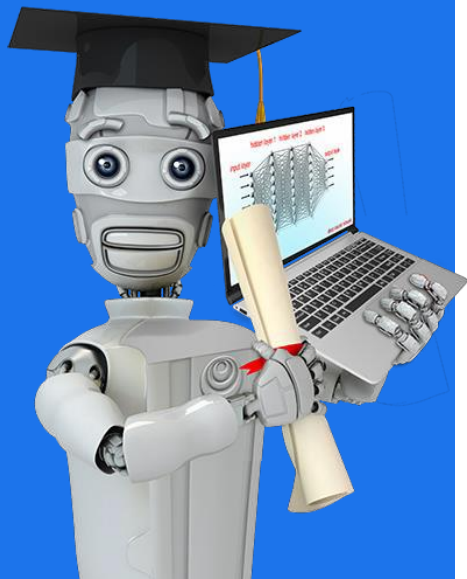
$$f_{\vec{w},b}(\vec{x}) = P(y = 1 | \vec{x}; \vec{w}, b)$$

Probability that y is 1,
given input $\vec{x}$, parameters $\vec{w}, b$

$$P(y = 0) + P(y = 1) = 1$$

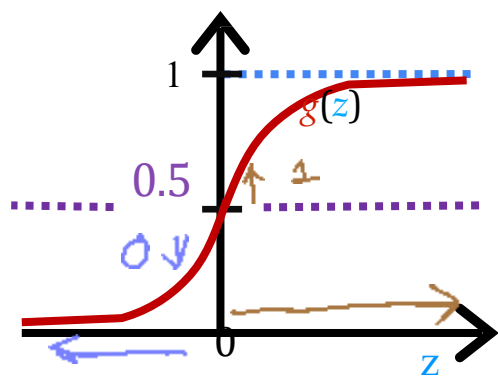
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Classification

Decision Boundary



$$\begin{aligned} \hat{y}_{\vec{w}, b}(\vec{x}) &= g(\underbrace{\vec{w} \cdot \vec{x} + b}_z) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}} \\ &= \mathcal{E}(y = 1 \mid \vec{w}, b) \quad 0.7 \quad 0.3 \\ &\quad \text{0 or 1?} \quad \text{threshold} \end{aligned}$$

Is $\hat{y}_{\vec{w}, b}(\vec{x}) \geq 0.5$?

Yes: $\hat{y} = 1$

No: $\hat{y} = 0$

When is

$$\hat{y}_{\vec{w}, b}(\vec{x}) \geq 0.5 \iff g(z) \geq 0.5$$

$$z \geq 0$$

$$< 0$$

$$\vec{w} \cdot \vec{x} + b \geq 0$$

$$\vec{w} \cdot \vec{x} + b < 0$$

$$\hat{y} = 1$$

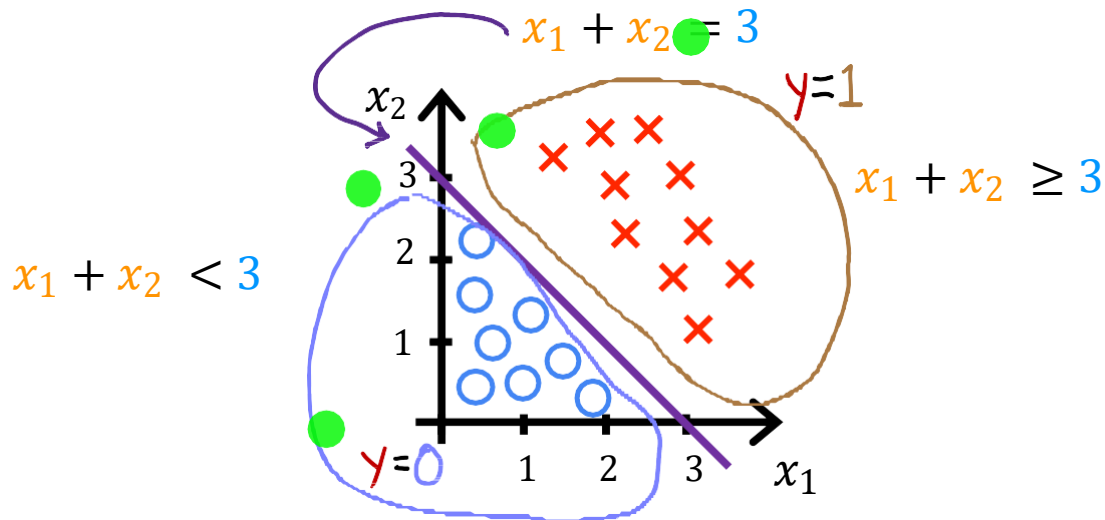
$$\hat{y} = 0$$

$$\begin{aligned} \hat{y}_{\vec{w}, b}(\vec{x}) & \\ \boxed{z = \vec{w} \cdot \vec{x} + b} & \\ \downarrow z & \\ \boxed{g(z) = \frac{1}{1 + e^{-z}}} & \end{aligned}$$

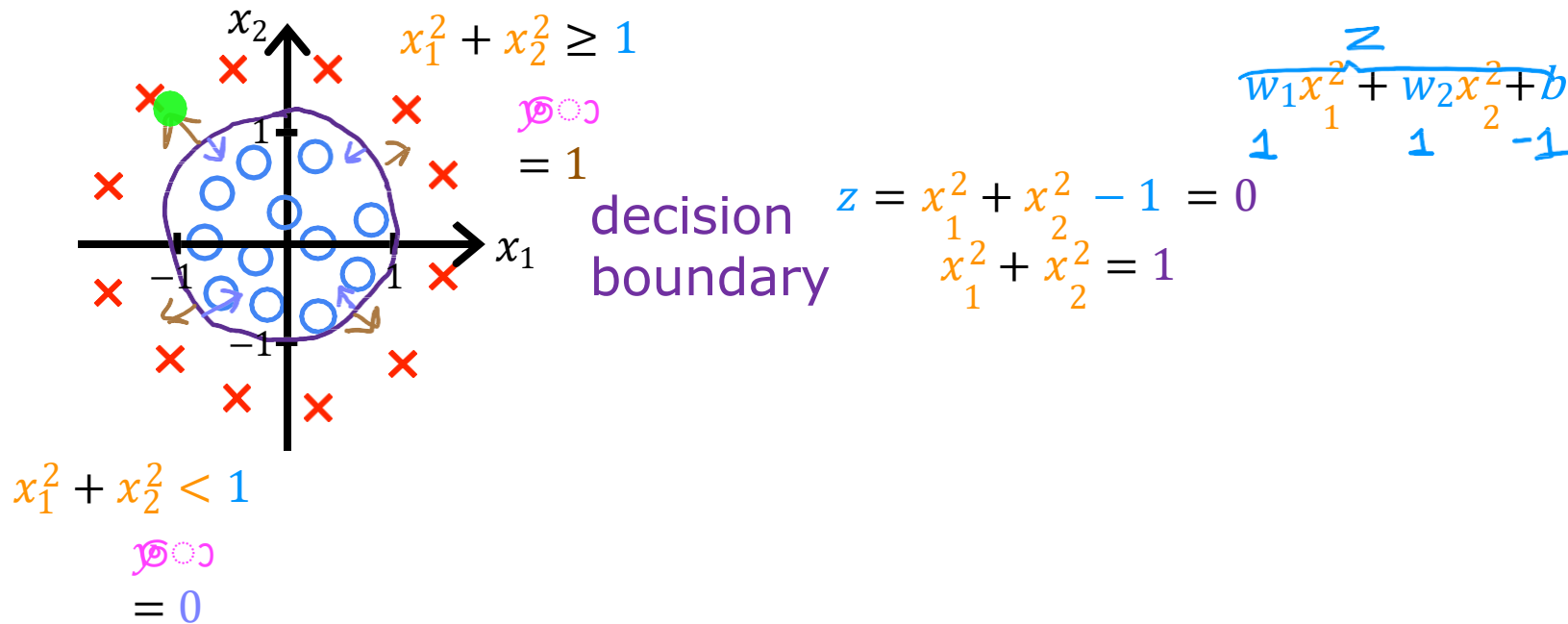
Decision boundary

$$f_{\vec{w},b}(\vec{x}) = g(z) = g(\underbrace{w_1x_1 + w_2x_2 + b}_{z})$$

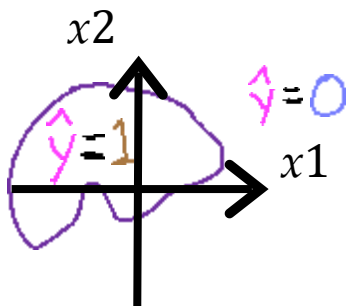
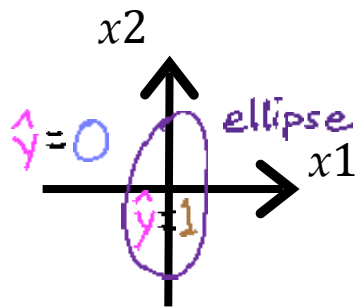
Decision boundary $z = \vec{w} \cdot \vec{x} + b = 0$
 $z = x_1 + x_2 - 3 = 0$



Non-linear decision boundaries



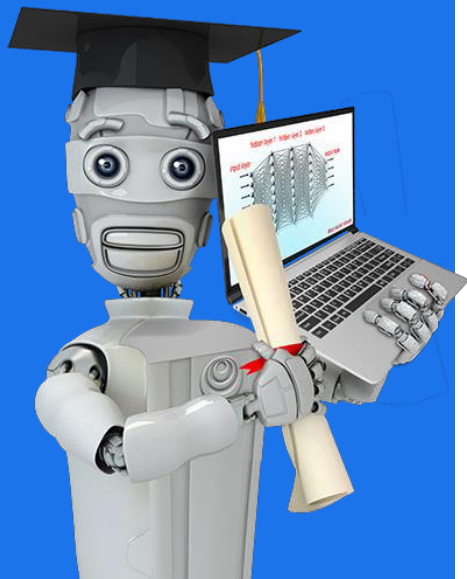
Non-linear decision boundaries



$$J_{\vec{w}, b}(\vec{x}) = g(z) = g(w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_1x_2 + w_5x_2^2 + w_6x_1^3 + \dots + b)$$

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Cost Function

Cost Function for Logistic Regression

Training set

	tumor size (cm) x_1	...	patient's age x_n	malignant? y	$i = 1, \dots, m \leftarrow$ training examples $j = 1, \dots, n \leftarrow$ features
$i=1$	10		52	1	target y is 0 or 1 $f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$
$\vdots$	2		73	0	
$\vdots$	5		55	0	
$\vdots$	12		49	1	
$i=m$	...		...	...	

How to choose $\vec{w} = [w_1 \ w_2 \ \dots \ w_n]$ and b ?

Squared error cost

cost

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2$$

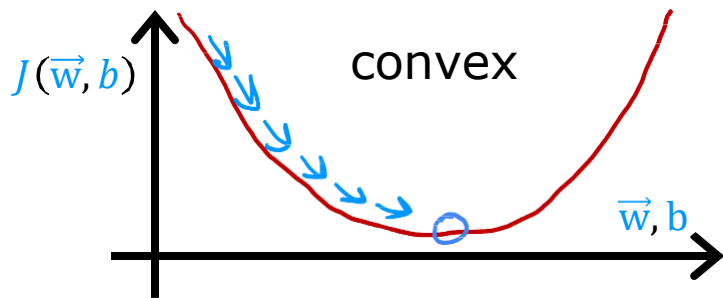
average of training set

loss

$$L(f_{\vec{w}, b}(\vec{x}^{(i)}), y^{(i)})$$

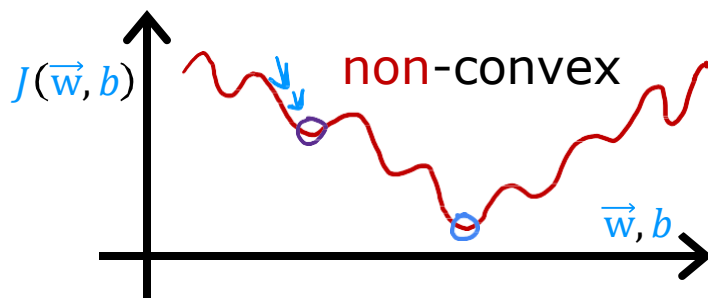
linear regression

$$f_{\vec{w}, b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$



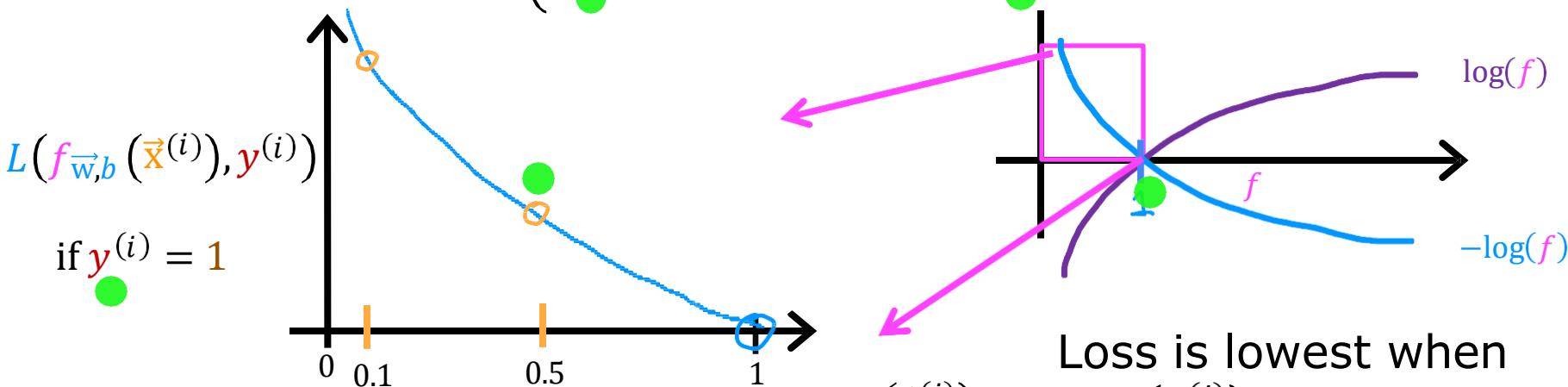
logistic regression

$$f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$



Logistic loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$



As $f_{\vec{w},b}(\vec{x}^{(i)}) \rightarrow 1$ then loss $\rightarrow 0$

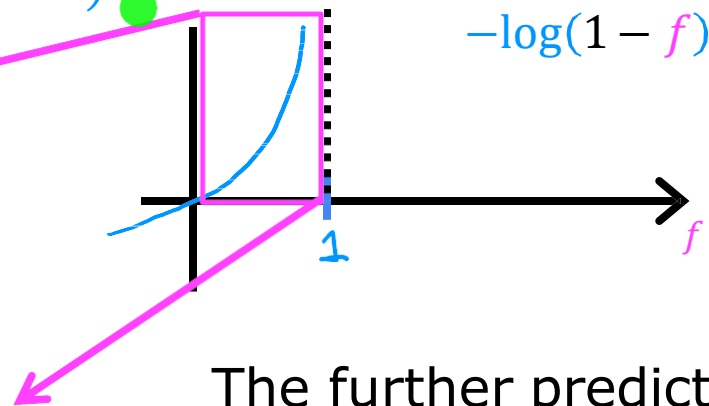
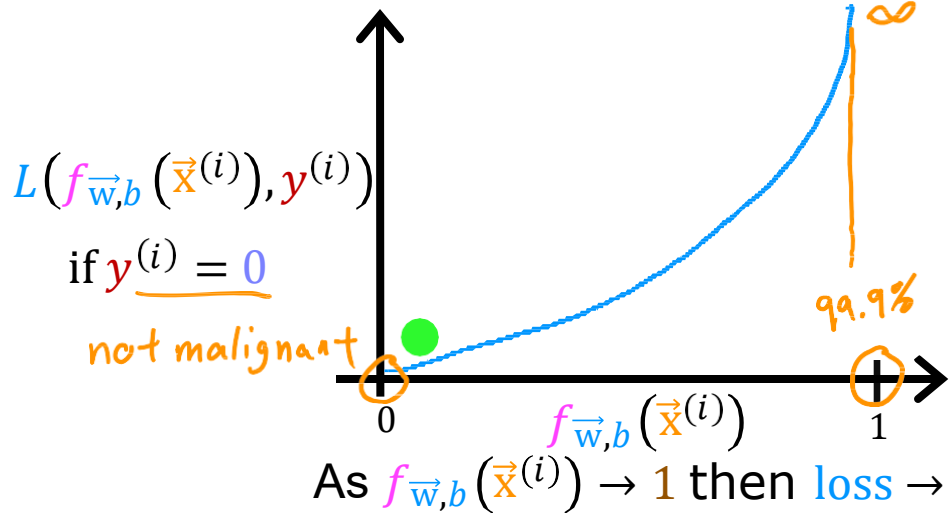
As $f_{\vec{w},b}(\vec{x}^{(i)}) \rightarrow 0$ then loss $\rightarrow \infty$

Loss is lowest when $f_{\vec{w},b}(\vec{x}^{(i)})$ predicts close to true label $y^{(i)}$.

Logistic loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

As $f_{\vec{w},b}(\vec{x}^{(i)}) \rightarrow 0$ then $\text{loss} \rightarrow 0$ $\Downarrow$



The further prediction $f_{\vec{w},b}(\vec{x}^{(i)})$ is from target $y^{(i)}$, the higher the loss.

Cost

$$\text{cost } J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m \underbrace{L(f_{\vec{w}, b}(\vec{x}^{(i)}), y^{(i)})}_{\text{loss}}$$

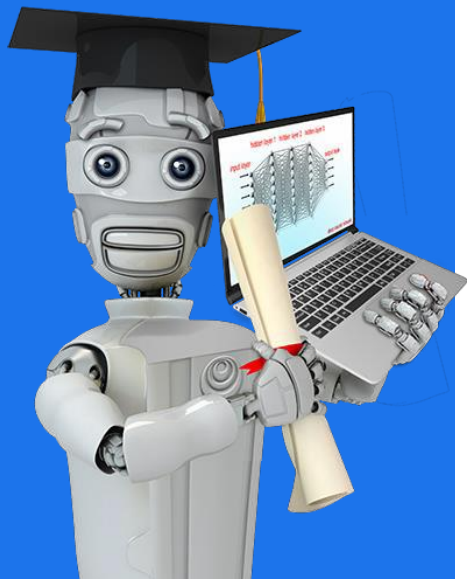
$$= \begin{cases} -\log(f_{\vec{w}, b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

if $y^{(i)} = 1$ *convex* $\rightarrow$ can reach a global minimum
if $y^{(i)} = 0$ *convex* $\rightarrow$ can reach a global minimum

find w, b that minimize cost J

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Cost Function

Simplified Cost Function for Logistic Regression

Simplified loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) - (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))$$

if $y^{(i)} = 1$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -1 \log(f_{\vec{w},b}(\vec{x}^{(i)}))$$

Simplified loss function

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = - \underbrace{y^{(i)}}_0 \log(\cancel{f_{\vec{w},b}(\vec{x}^{(i)})}) - \underbrace{(1 - y^{(i)})}_{(1-0)} \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))$$

if $y^{(i)} = 1$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = -1 \log(f(\hat{x}))$$

if $y^{(i)} = 0$:

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \underbrace{-(1-0) \log(1 - f(\hat{x}))}$$

Simplified cost function

loss

$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = - \underbrace{y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))}_{\text{convex (single global minimum)}}$$

cost

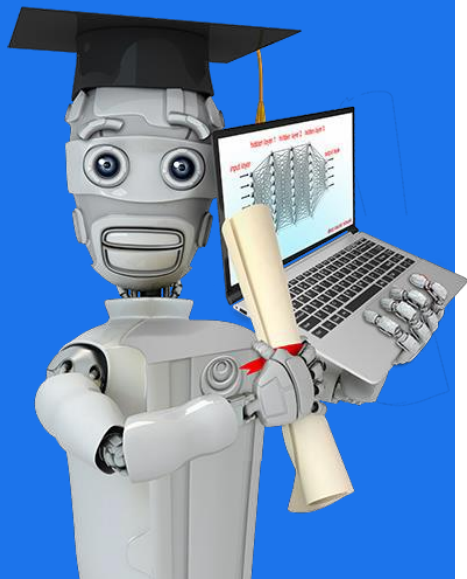
$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m [L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)})]$$

$$= \frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)}))]$$

maximum likelihood
(don't worry about it!)

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Gradient Descent

Gradient Descent Implementation

Training logistic regression

Find $\vec{w}, b$

Given new $\vec{x}$, output $f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$

$$P(y = 1 | \vec{x}; \vec{w}, b)$$

Gradient descent

cost

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w}, b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) \right]$$

repeat {

$j=1 \dots n$

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

} simultaneous updates

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$\frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})$$

Gradient descent for logistic regression

repeat {

looks like linear regression!

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{X}^{(i)}) - y^{(i)}) x_j^{(i)} \right]$$

$$b = b - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{X}^{(i)}) - y^{(i)}) \right]$$

} simultaneous updates

Same concepts:

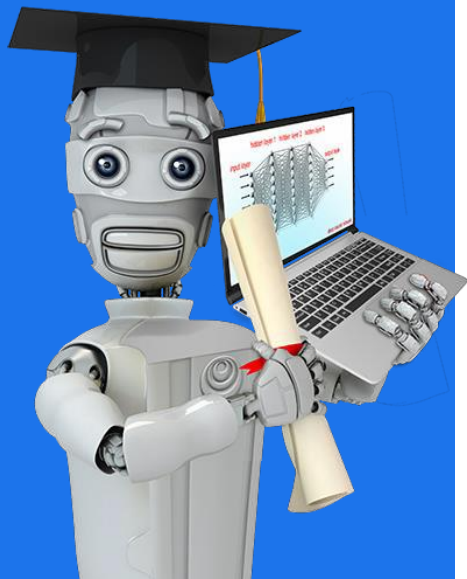
- Monitor gradient descent (learning curve)
- Vectorized implementation
- Feature scaling

Linear regression $f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

Logistic regression $f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{(-\vec{w} \cdot \vec{x} + b)}}$

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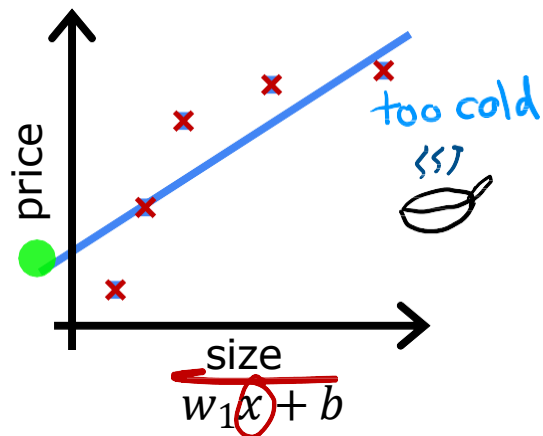
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Regularization to Reduce Overfitting

The Problem of Overfitting

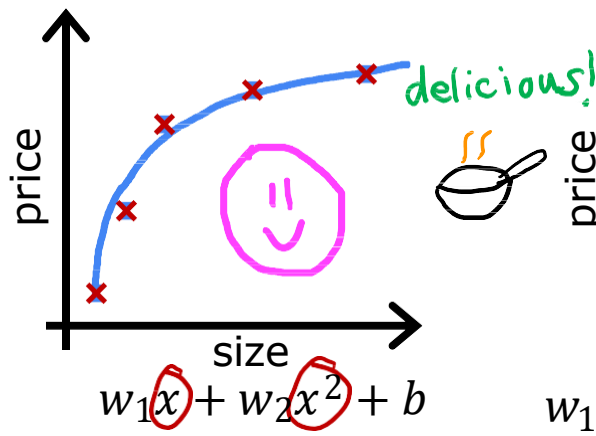
Regression example



underfit

- Does not fit the training set well

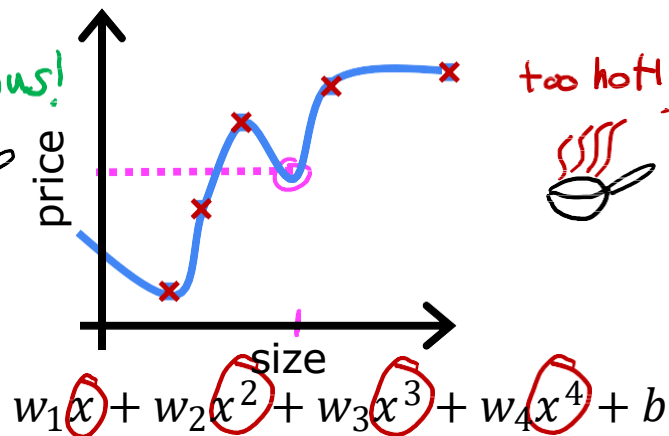
high bias



just right

- Fits training set pretty well

generalization

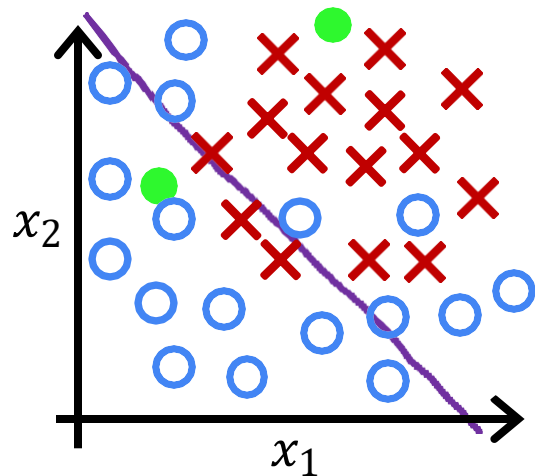


overfit

- Fits the training set extremely well

high variance

Classification

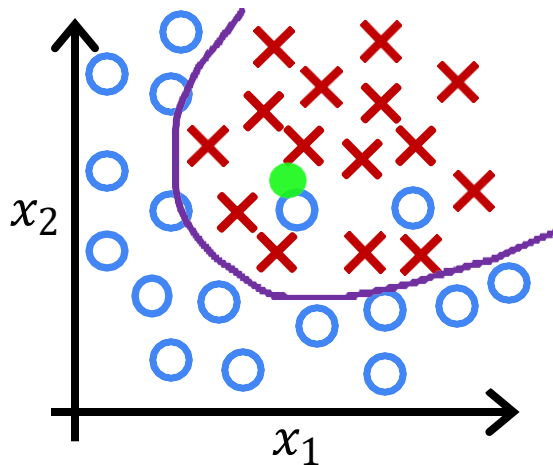


$$z = w_1 x_1 + w_2 x_2 + b$$

$$f_{\vec{w}, b}(\vec{x}) = g(z)$$

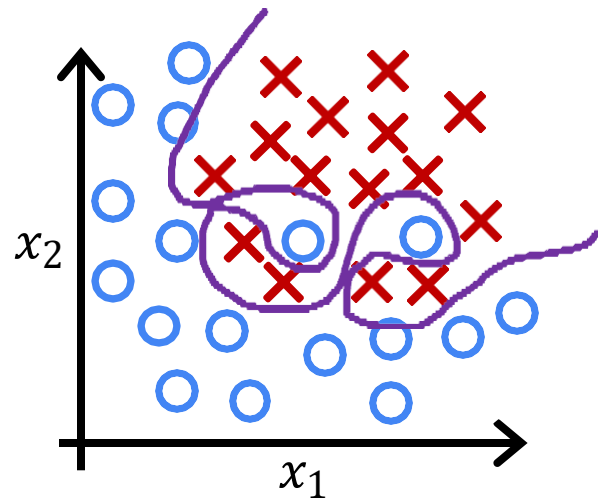
g is the sigmoid function

underfit high bias



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + b$$

just right

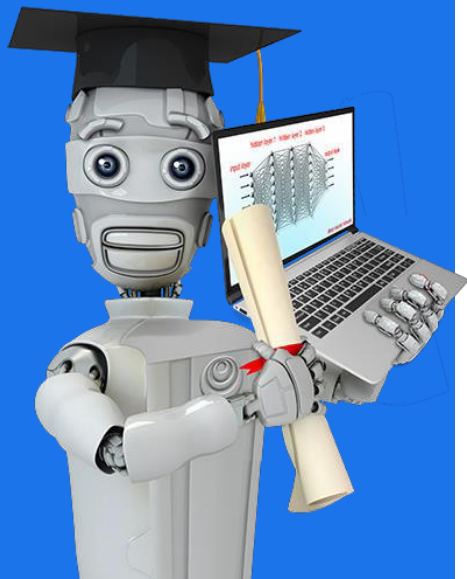


$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + w_6 x_1^3 x_2 + \dots + b$$

overfit

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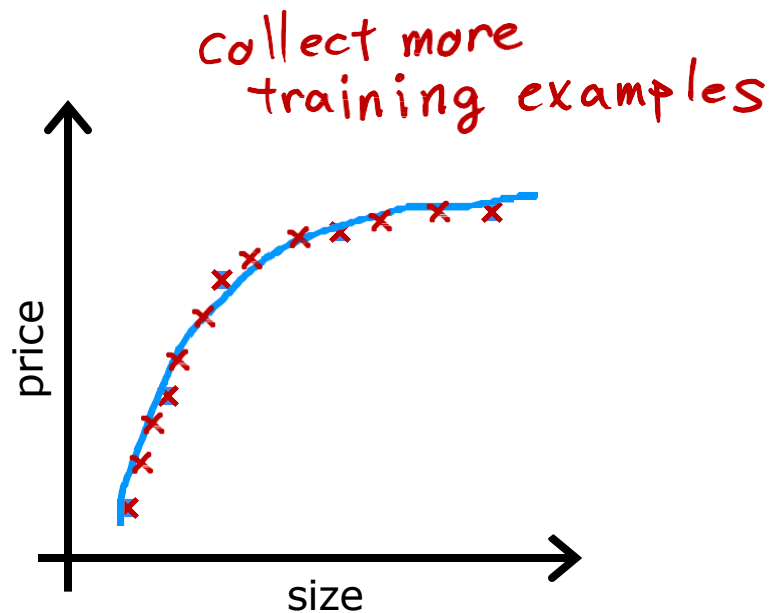
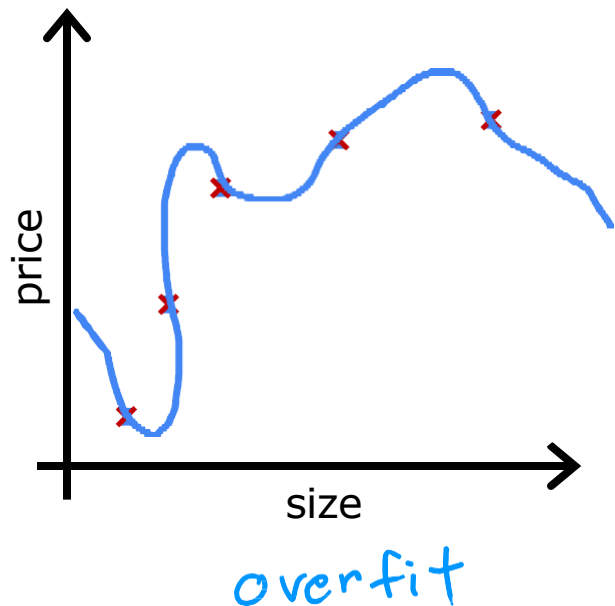
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Regularization to Reduce Overfitting

Addressing Overfitting

Collect more training examples



Select features to include/exclude

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y
							

all features



insufficient data



overfit

selected features

size
bedrooms
age
just right
feature selection

course 2

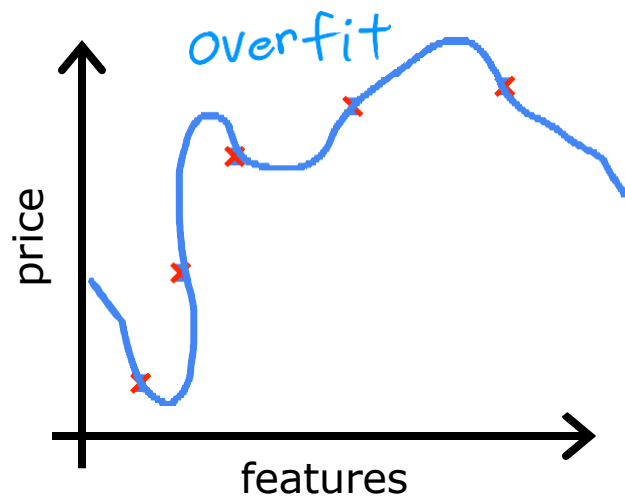
disadvantage



useful features
could be lost

Regularization

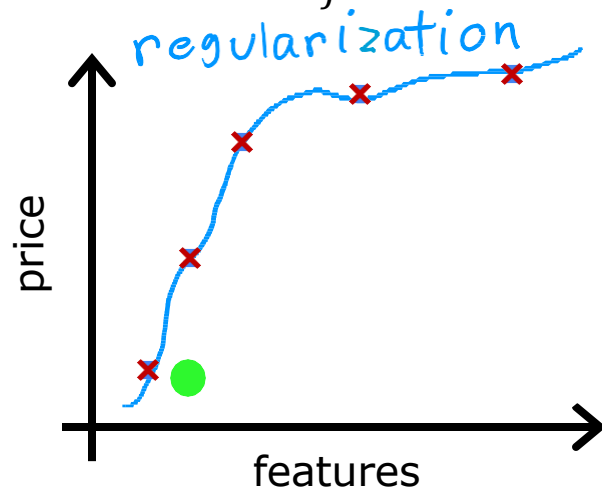
Reduce the size of parameters w_j



$$f(x) = 28x - 385x^2 + 39x^3 - 174x^4 + 100$$

large values for w_j

eliminate feature



$$f(x) = 13x - 0.23x^2 + 0.000014x^3 - 0.0001x^4 + 10$$

small values for w_j

Addressing overfitting

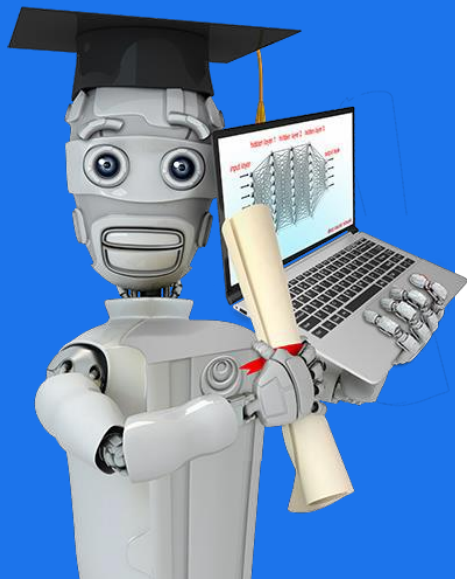
Options

1. Collect more data
2. Select features
 - Feature selection *in course 2*
3. Reduce size of parameters
 - “Regularization” *next videos!*



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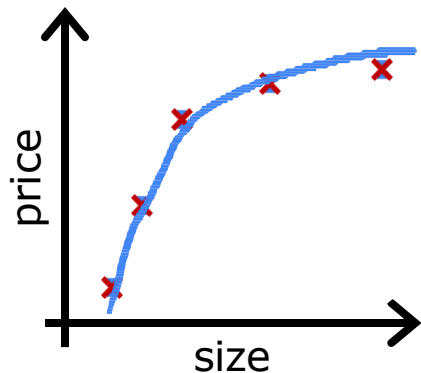
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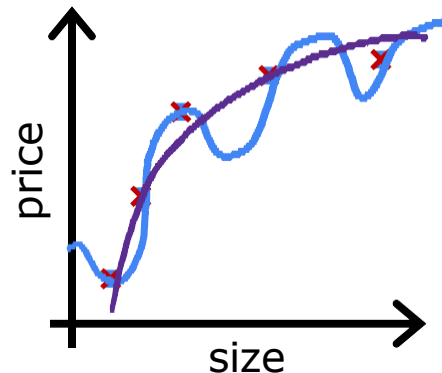
Regularization to Reduce Overfitting

Cost Function with Regularization

Intuition



$$w_1x + w_2x^2 + b$$



$$w_1x + w_2x^2 + \underbrace{w_3x^3}_{\approx 0} + \underbrace{w_4x^4}_{\approx 0} + b$$

make w_3, w_4 really small (≈ 0)

$$\min_{\vec{w}, b} \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2$$

$$+ 1000 \underbrace{w_3^2}_{0.001} + 1000 \underbrace{w_4^2}_{0.002}$$

Regularization

small values $w_1, w_2, \dots, w_n, b$

simpler model

less likely to overfit

$$w_3 \approx 0$$

$$w_4 \approx 0$$

size x_1	bedrooms x_2	floors x_3	age x_4	avg income x_5	...	distance to coffee shop x_{100}	price y
---------------	-------------------	-----------------	--------------	------------------------	-----	---	--------------

$$w_1, w_1, w_2, \dots, w_{100}, b$$

n features

$$n = 100$$

$$J(\vec{w}, b) = \frac{1}{2m} \left[\sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{"lambda" regularization parameter}} + \underbrace{\frac{\lambda}{2m} b^2}_{\text{can include or exclude } b} \right]$$

$\lambda > 0$

Regularization

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left[\underbrace{\frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2}_{\text{mean squared error}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}} \right]$$

fit data

Keep w_j small

λ balances both goals

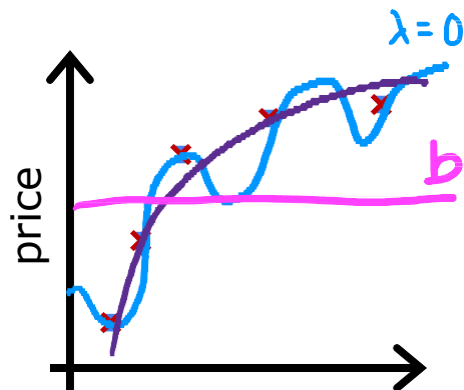
choose $\lambda = 10^{10}$

$$f_{\vec{w}, b}(\vec{x}) = \cancel{w_1}x + \cancel{w_2}x^2 + \cancel{w_3}x^3 + \cancel{w_4}x^4 + b$$

≈ 0 ≈ 0 ≈ 0 ≈ 0

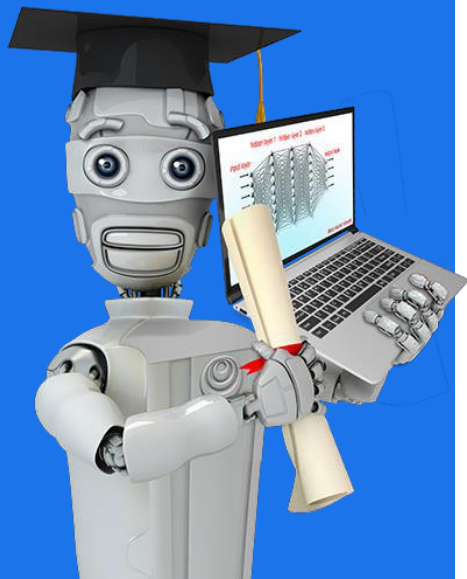
$$f(x) = b$$

Choose λ



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Regularization to Reduce Overfitting

Regularized Linear Regression

Regularized linear regression

$$\min_{\vec{w}, b} J(\vec{w}, b) = \min_{\vec{w}, b} \left[\frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right]$$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} w_j$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})$$

don't have to regularize b

} simultaneous update

Implementing gradient descent

repeat {

- $w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})$$

} simultaneous update $j = 1 \dots n$

Implementing gradient descent

repeat {

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})$$

} simultaneous update $j = 1 \dots n$

$$w_j = \underbrace{w_j - \alpha \frac{\lambda}{m} w_j}_{w_j \left(1 - \alpha \frac{\lambda}{m} \right)} - \underbrace{\alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}}_{\text{usual update}}$$

shrink w_j

$$\alpha \frac{\lambda}{m} = 0.01 \frac{1}{50} = 0.0002$$

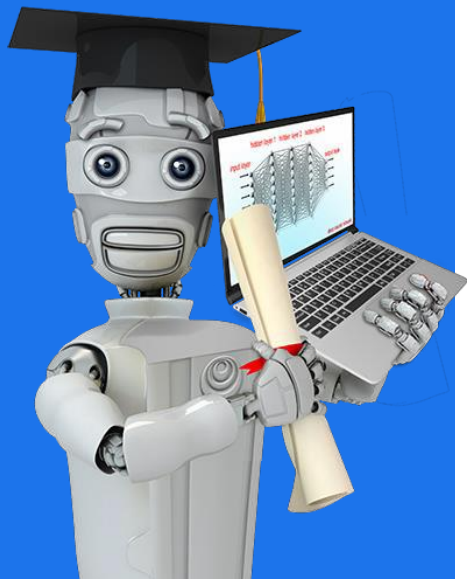
$$w_j (1 - 0.0002) = 0.9998 w_j$$

How we get the derivative term (optional)

$$\begin{aligned}
 \bullet \frac{\partial}{\partial w_j} J(\vec{w}, b) &= \frac{d}{dw_j} \left[\frac{1}{2m} \sum_{i=1}^m \underbrace{(f(\vec{x}^{(i)}) - y^{(i)})^2}_{\vec{w} \cdot \vec{x}^{(i)} + b} + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right] \\
 &= \cancel{\frac{1}{2m} \sum_{i=1}^m} \left[(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)}) \cancel{2} x_j^{(i)} \right] + \cancel{\frac{\lambda}{2m}} \cancel{2} w_j \quad \text{No } \sum_{j=1}^n \\
 &= \frac{1}{m} \sum_{i=1}^m \left[\underbrace{(\vec{w} \cdot \vec{x}^{(i)} + b - y^{(i)})}_{f(\vec{x})} x_j^{(i)} \right] + \frac{\lambda}{m} w_j \\
 \bullet &= \frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j
 \end{aligned}$$

Stanford
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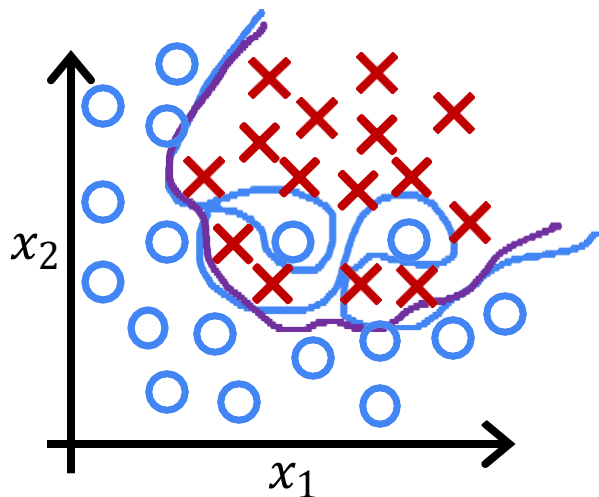
DeepLearning.AI



Regularization to Reduce Overfitting

Regularized Logistic Regression

Regularized logistic regression



$$z = w_1x_1 + w_2x_2 + w_3x_1^2x_2 + w_4x_1^2x_2^2 + w_5x_1^2x_2^3 + \dots + b$$

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-z}}$$

Cost function

$$J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w},b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

$\min_{\vec{w}, b} J(\vec{w}, b) \rightarrow w_j \downarrow$

Regularized logistic regression

$$J(\vec{w}, b) = \min_{\vec{w}, b} \left[-\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(f_{\vec{w}, b}(\vec{x}^{(i)})) + (1 - y^{(i)}) \log(1 - f_{\vec{w}, b}(\vec{x}^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2 \right]$$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$j = 1 \dots n$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

}

Looks same as
for linear regression!

$$= \frac{1}{m} \sum_{i=1}^m \left[(f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right] + \frac{\lambda}{m} w_j$$

logistic regression

don't have to
regularize b